Deep Hucks Or Small Ball?: Spatial Patterns and Possession Value Among Four 2026 Semifinalists in the Ultimate Frisbee Association

Theodore Taussig

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Abstract

Professional ultimate possessions are often summarized by their endpoint: a goal, a turnover, or a break. That summary leaves out the route. This paper examines the repeated spatial routes used by the New York Empire, Austin Sol, Minnesota Wind Chill, and Oakland Spiders during the 2026 regular season. We combine throw-level Shown Space data with manually curated field-path groups, then describe each selected pattern using frequency, goal/turnover efficiency, total aEC per possession, aEC per throw, and throw count. The result is a team-centered view of offensive identity. The same outcome can emerge from a short, aggressive sequence, a longer reset-heavy possession, or a low-huck path that steadily compresses the field. A league-wide throw-count heatmap provides the broader context: five throws is the modal length of an O-line goal possession, but the four teams examined here occupy noticeably different parts of that distribution.

1 Introduction

The modern Ultimate Frisbee Association (UFA) analytics conversation has moved well beyond the final line in a box score. Shown Space was introduced as a way to use the league's play-by-play infrastructure to ask how possessions develop, which actions change scoring probability, and which players drive that change (Eberhard, 2026d). That shift matters because two possessions can finish in the same end zone while asking very different questions of an offense. One may win through a full field deep throw; another may survive several resets before a cut finally opens the goal line.

The Sloan work behind Shown Space established a related foundation by combining machine learning with completion probability and field value to estimate player decision-making in professional ultimate (Eberhard et al., 2025). The public articles that followed use that foundation as a storytelling device. In the six-passer comparison, for example, aEC is paired with complementary measures so that distinct jobs remain visible rather than being flattened into one ranking (Eberhard, 2026c). The lesson transfers naturally from players to teams: a leaderboard can identify who is productive, while paths can help explain what that productivity looks like.

This paper applies that idea to the four UFA semifinalists represented in the current possession browser. The question is deliberately narrower than "Which team is best?" We ask: *What are the recurring spatial forms of each team's offense, and how do their outcomes and aEC values differ?* The answer should not be reduced to a single style label. The UFA's own team analysis emphasizes that the O-line and D-line can become meaningfully different teams after the disc changes hands, and that a deep ball is a tool rather than a complete offensive identity (Eberhard, 2026e). Our analysis therefore stays with O-line possessions, includes both goals and turnovers, and treats spatial organization as evidence to be inspected rather than a substitute for judgment.

2 Data and Methods

2.1 Sample and possession reconstruction

The analysis uses the cached per-game Shown Space throw files in this project. The regular-season window runs from April 24 through July 19, 2026. The cache also contains later playoff files, so the figure-building script excludes games dated after July 19 before calculating any count, efficiency, or aEC statistic. Each focus team contributes 12 regular-season games. The resulting reconstructed O-line samples are 257 possessions for New York, 285 for Austin, 266 for Minnesota, and 266 for Oakland.

A possession is formed by grouping throws with the same game, quarter, point, possession number, and home/away side. The final throw determines the outcome: the two retained outcomes are 'goal' and 'turnover'. The line type comes from the source `o_line` field. For every possession, the pipeline retains the ordered throw path, starting and ending field coordinates, throw count, field progress, huck count, reset count, throw-level aEC, and the derived possession totals.

The word "efficiency" has a deliberately local meaning in this paper. For a group of reconstructed O-line possessions, it is

$$\text{efficiency} = \frac{\text{goals}}{\text{goals} + \text{turnovers}}.$$

It is therefore an outcome ratio for this sample, not a claim that it is identical to the UFA's published OE or expectation-adjusted OEOE measures. Keeping that distinction explicit is essential when comparing a custom browser view with a public leaderboard.

2.2 Human-in-the-loop spatial grouping

The first three rows in each saved arrangement checkpoint (`data/arrangements/2026/empire-paper.json`, `sol-paper.json`, `windchill-paper.json`, and `spiders-paper.json`) are treated as that team's three selected recurring patterns. The grouping was made by reviewing field cards and overlaying paths, then deciding which routes were similar enough to share a row. This is a manual taxonomy with a reproducible checkpoint, not an unsupervised claim that the game contains exactly three natural clusters.

For the paper, a card contributes to a pattern's statistics only when its possession is in the regular-season data and its line type is O-line. The field figures draw every retained path in a selected group, with goals in blue and turnovers in red. That choice keeps the display honest: a group can look spatially coherent while still containing different outcomes.

2.3 Metrics

For a pattern g with n_g possessions and T_g total throws, let $A_i = \sum_{j=1}^{T_i} aEC_{ij}$ denote the total aEC of possession i. We report:

- **frequency**: n_g/N_t, where N_t is the team's regular-season O-line possession count;
- **efficiency**: the goal fraction defined above, with turnovers in the denominator;
- **total aEC per possession**: $\frac{1}{n_g}\sum_{i\in g} A_i$; this is the arithmetic mean of the possession-level total aEC values in the selected pattern;
- **aEC per throw**: $\frac{\sum_{i\in g} A_i}{\sum_{i\in g} T_i}$; this is the pattern's total aEC divided by its total throws; and
- **median throws**: the median number of throws in the pattern.

The throw-level aEC values are taken from the cached Shown Space rows and summed within each possession. Thus, yes, aEC per possession for a pattern is the aEC from all retained possessions in that pattern divided by the number of retained possessions. It is a group mean, not a raw team total. By contrast, aEC per throw weights the pattern by its throws rather than giving every possession equal weight. Reporting both measures prevents a short possession from being mistaken for a high-volume one.

2.4 Why aEC can be above one

The Sloan paper's normalized drive-level illustration is useful for intuition: one complete scoring drive can be represented as one unit of contribution (Eberhard et al., 2025). In this project, however, the browser consumes the cached per-throw aEC values and adds them; it does not clip each possession to the interval $[-1, 1]$, force every goal to exactly one, or renormalize the selected pattern after the fact. The resulting total is therefore an additive contribution score, not a probability and not a hard upper-bounded scoring total. A goal possession can consequently be slightly above one, while a turnover can be negative. Those values reflect the model output and the aggregation convention used here; they should be compared consistently across possessions rather than read as literal probabilities of scoring.

3 Results

3.1 The league context: five throws is common, not destiny

Figure 1 shows the throw-count distribution for regular-season O-line goal possessions across all 22 cached teams. Each row is normalized within team, so the darkest cell answers a within-team question: which throw count appears most often among that team's goals? Across the league, the mode is five throws among 3,607 O-line goal possessions. The focus teams do not share that mode. Austin peaks at four throws; Oakland and Minnesota peak at eight; New York peaks at nine. A mode describes the most frequent length, not an offensive instruction. It is best read alongside the rest of the row, especially when two nearby counts have similar mass.

3.2 Team-level baseline

The four teams separate before the individual patterns are considered (Table 1). New York's reconstructed O-line converted 180 of 257 possessions, or 70.0 percent. Oakland followed at 63.5 percent, while Minnesota and Austin finished at 61.3 and 60.0 percent, respectively. New York also led this local sample in average total aEC per possession (0.592) and aggregate aEC per throw (0.070). Those values should be interpreted as descriptive summaries of the cached rows, not as a replacement for the public Shown Space team metrics.

Table 1: Regular-season reconstructed O-line baseline.

Team	Games	Pos.	Goals	TOs	Eff.	aEC/pos.	aEC/throw
New York Empire	12	257	180	77	70.0%	0.592	
Austin Sol	12	285	171	114	60.0%	0.431	
Minnesota Wind Chill	12	266	163	103	61.3%	0.486	
Oakland Spiders	12	266	169	97	63.5%	0.492	

3.3 New York Empire: a direct branch beside two mixed routes

The new Empire checkpoint makes the largest selected group Pattern 1, saved as "O-line pattern 22." It contains 33 regular-season possessions: 27 goals and six turnovers, or 81.8 percent efficiency. Its median

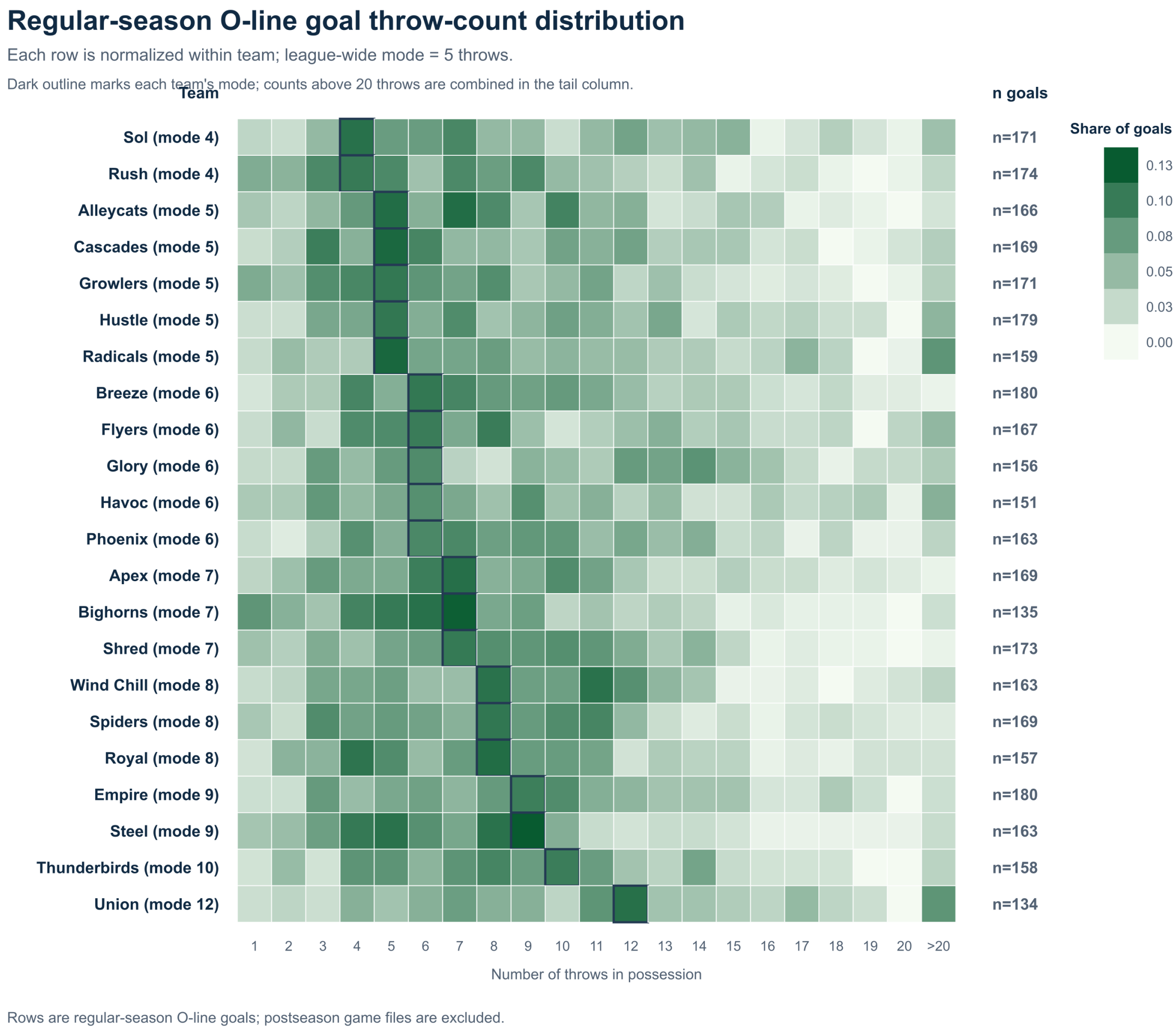

Figure 1: Regular-season throw-count distribution for O-line goal possessions. Color is the share of each team's O-line goals, not the share of all league possessions. The right-hand count is the number of goal possessions used for that row.

length is eight throws, and 42.4 percent of the possessions include a huck. Pattern 2, saved as "pattern 07," is much smaller at 20 possessions, but its direct three-throw shape is distinctive: every retained possession includes a huck, and its aggregate aEC per throw is the highest of the Empire rows at 0.190. Pattern 3, "pattern 23," contains 11 possessions and has the same 81.8 percent efficiency as Pattern 1.

The value measures add an important qualification. Pattern 2's short sequence produces 0.702 mean total aEC per possession, whereas Pattern 3 reaches 0.827 despite taking a median of eight throws; Pattern 1 sits between them at 0.796. Empire's selected routes therefore illustrate two different kinds of value density: a compact, aggressive branch and longer mixed paths that accumulate more value across the possession. The similar conversion rates make the spatial contrast more informative than a simple success ranking.

3.4 Austin Sol: long mixed routes with a shorter aggressive branch

Sol's first selected group, "pattern 28," is the largest: 25 possessions, 20 goals, and five turnovers, for 80.0 percent efficiency. It is also the longest of the three by median length at 13 throws, although only 44.0 percent of its paths contain a huck. Pattern 2, "pattern 25," contributes 13 possessions at 76.9 percent efficiency, with a 10-throw median and the lowest median starting position of the three. Pattern 3, "pattern 12," is the smallest at 12 possessions; it has a five-throw median and a huck in 91.7 percent of its paths.

The aEC measures separate these routes cleanly. Patterns 1 and 2 have nearly identical mean totals, 0.762 and 0.767, while their aggregate aEC per throw is only 0.060 and 0.077 because the sequences are long. Pattern 3 reaches the highest density, 0.124 per throw, but its mean total is lower at 0.649 and its efficiency is 75.0 percent. Sol's selected set thus pairs a high-volume, mixed progression with a shorter deep option; the latter is efficient in value per throw without dominating the complete-possession measure.

3.5 Minnesota Wind Chill: a short direct route inside a mixed set

Wind Chill's new Pattern 1, saved as "pattern 15," includes 23 possessions, 16 goals, and seven turnovers. Its 69.6 percent efficiency, eight-throw median, and 0.642 mean total aEC describe a substantial but uneven route family. Pattern 2, "pattern 09," is slightly more frequent at 24 possessions and more successful at 79.2 percent. Its direct four-to-five-throw shape includes a huck in 87.5 percent of cases and produces the strongest density of the three, 0.141 aEC per throw. Pattern 3, "pattern 23," contains 16 possessions, has a 10.5-throw median, and converts 68.8 percent.

The selected Wind Chill paths therefore do not point to one fixed tempo. The short, huck-heavy branch is the most efficient in both outcome rate and aEC per throw, while the two longer mixed groups provide more of the visual volume. That distinction matters when the league-wide mode of five throws is used as a reference: Wind Chill has a common short branch, but its selected offense also contains longer ways to advance the disc.

3.6 Oakland Spiders: a high-volume branch and two efficient families

Oakland's new first group, "pattern 15," is the largest selected family at 29 possessions, but its 13 goals and 16 turnovers produce only 44.8 percent efficiency. It has a six-throw median and a huck in 65.5 percent of paths. The smaller second group, "pattern 18," looks very different: 14 possessions, 13 goals, one turnover, and 92.9 percent efficiency. It begins from an advanced median starting position of roughly 45 yards and uses a median of eight throws. Pattern 3, "pattern 21," is also small at 11 possessions, with 10 goals, one turnover, 90.9 percent efficiency, a nine-throw median, and no hucks.

The selected Spiders groups are consequently a warning against treating a large overlay as a representative one. Pattern 1 has the lowest mean total aEC, 0.348, and the lowest aEC per throw,

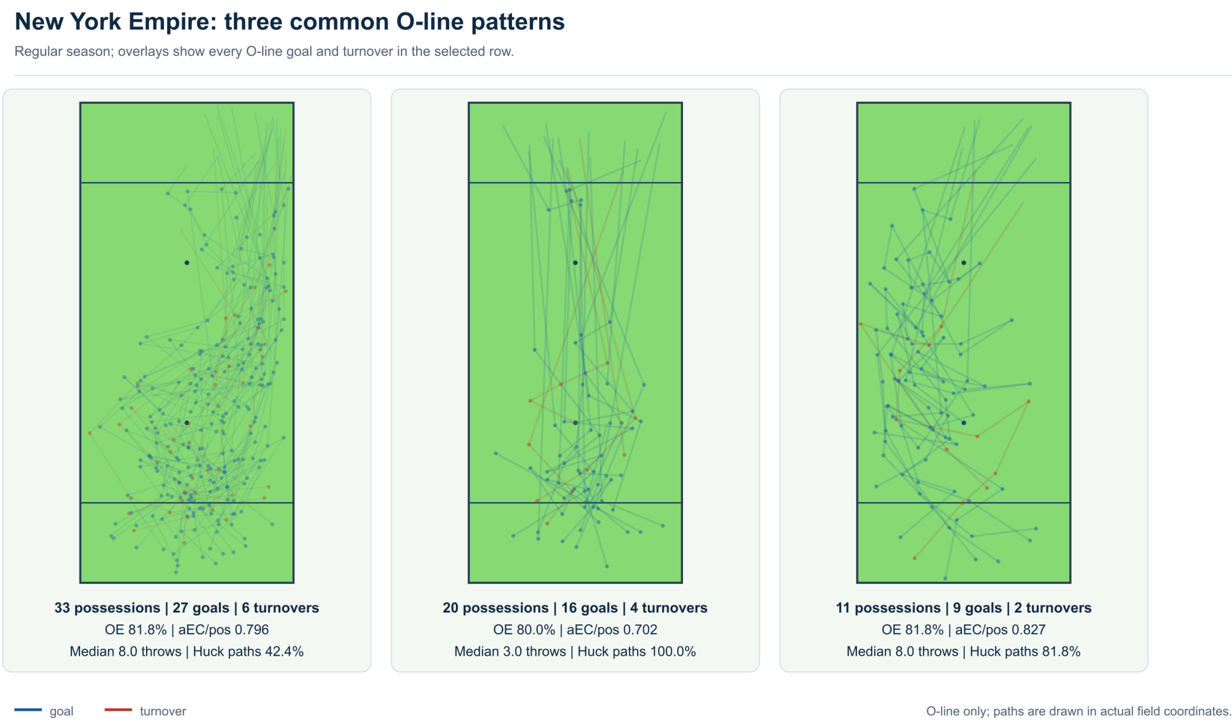

Figure 2: New York Empire's three selected recurring patterns. Each field overlays the regular-season O-line paths retained in that row; blue is a goal and red is a turnover. The metrics below each field include both outcomes.

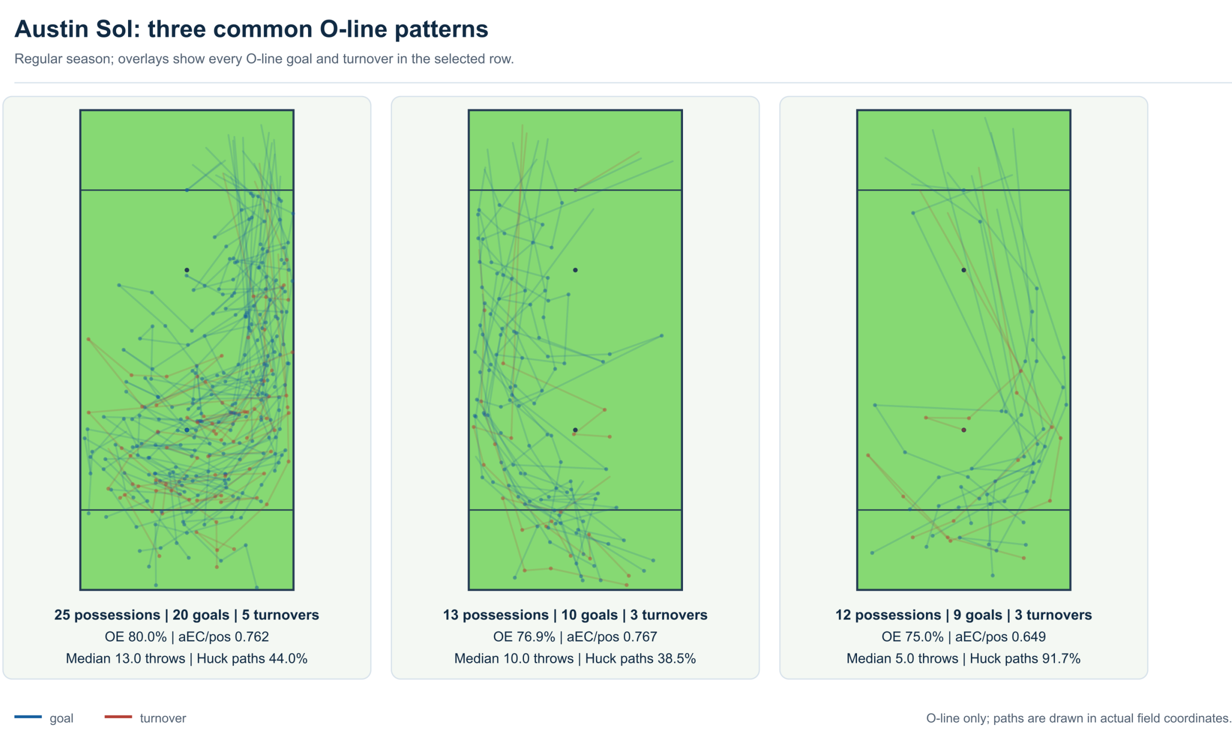

Figure 3: Austin Sol's three selected recurring patterns. The selected rows include both goal and turnover paths; the sample size beneath each field keeps the apparent 100 percent or near-100 percent rates in context.

0.056, while Patterns 2 and 3 are much stronger at 0.918 and 0.887 mean total aEC. Their paths support a more nuanced description: one common, huck-heavy family absorbs many turnovers, alongside two less frequent and much more efficient spatial families. The turnover-inclusive figures are what make that contrast visible.

3.7 The comparison in one table

Table 2 places the selected rows beside one another. The numbers show why frequency, efficiency, and aEC should be allowed to disagree. Empire's first row is its largest selected group, while the smaller second row has the highest aEC per throw among the Empire rows. Austin's shorter third row has the strongest value density, whereas its two longer groups carry more total value per possession. Wind Chill's direct branch combines the best efficiency and density in its selected set. Oakland has the sharpest contrast: its largest row is turnover-heavy, while the two smaller rows are highly efficient. Those comparisons are possible because the table keeps frequency, outcomes, and additive aEC side by side.

4 Discussion

4.1 What the patterns add to the leaderboard

The main contribution of these displays is interpretive rather than merely decorative. A possession path supplies a bridge between a metric and a tactical description. The six-passer article makes that bridge at the player level by showing that high aEC can come from very different combinations of shooting, volume, field vision, or support (Eberhard, 2026c). Here, the same logic operates one level up. Empire's selected rows suggest a direct aggressive branch alongside longer mixed solutions. Oakland's rows show why a team's most frequent family may not be its most efficient one. Austin and Minnesota both carry short, huck-heavy routes beside longer possessions that contribute more of the visual volume.

The distinction between total aEC and aEC per throw is especially useful. Total aEC asks how much value the entire possession accumulated. aEC per throw asks how concentrated that value was across the sequence. A short possession can look excellent on the latter measure even when another longer possession produces more total value. Neither is a universal quality score; each answers a different question about the route.

4.2 Turnovers change the pattern story

If only scoring possessions were retained, several of the selected overlays would look cleaner and several efficiency numbers would become meaningless. Including turnovers changes the interpretation in two ways. First, it makes "how often this pattern scores" an observable outcome ratio rather than a description of successful examples only. Second, it lets spatially similar routes reveal where the same idea breaks down. The red paths in the figures are not clutter; they are the negative evidence needed to distinguish a reliable pattern from a memorable one.

That is also why the manuscript reports n beside every percentage. A 100 percent group with two possessions is a promising observation. It is not a league-ready estimate. The public Shown Space writing repeatedly combines volume, efficiency, risk, and role when telling player stories (Eberhard, 2026b,a); team patterns deserve the same discipline.

4.3 From recurring paths to tactical language

The current labels should remain modest. "Lateral up left sideline" is a geometric description grounded in the cards. A claim such as "Oakland always attacks the left sideline" would require more evidence: force direction, opponent quality, personnel, game state, and the paths that did not score. Likewise, a

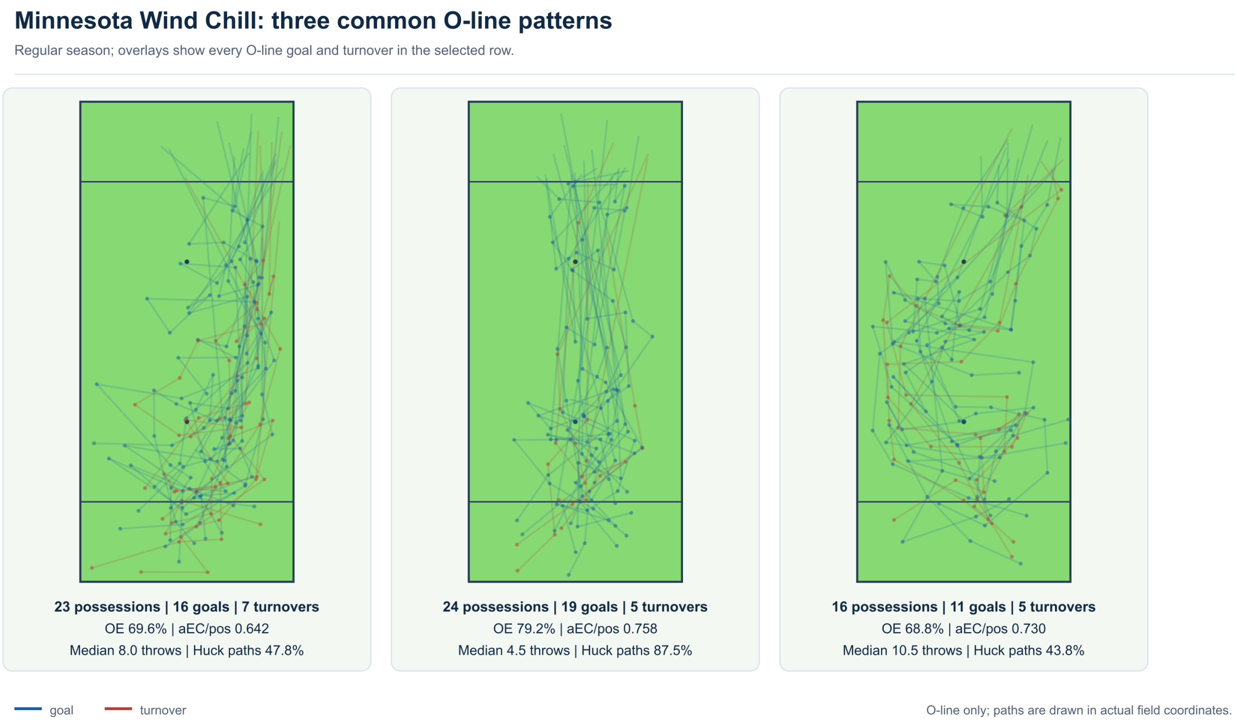

Figure 4: Minnesota Wind Chill's three selected recurring patterns. Pattern 3 is deliberately shown with its small n rather than being allowed to read as a general team tendency.

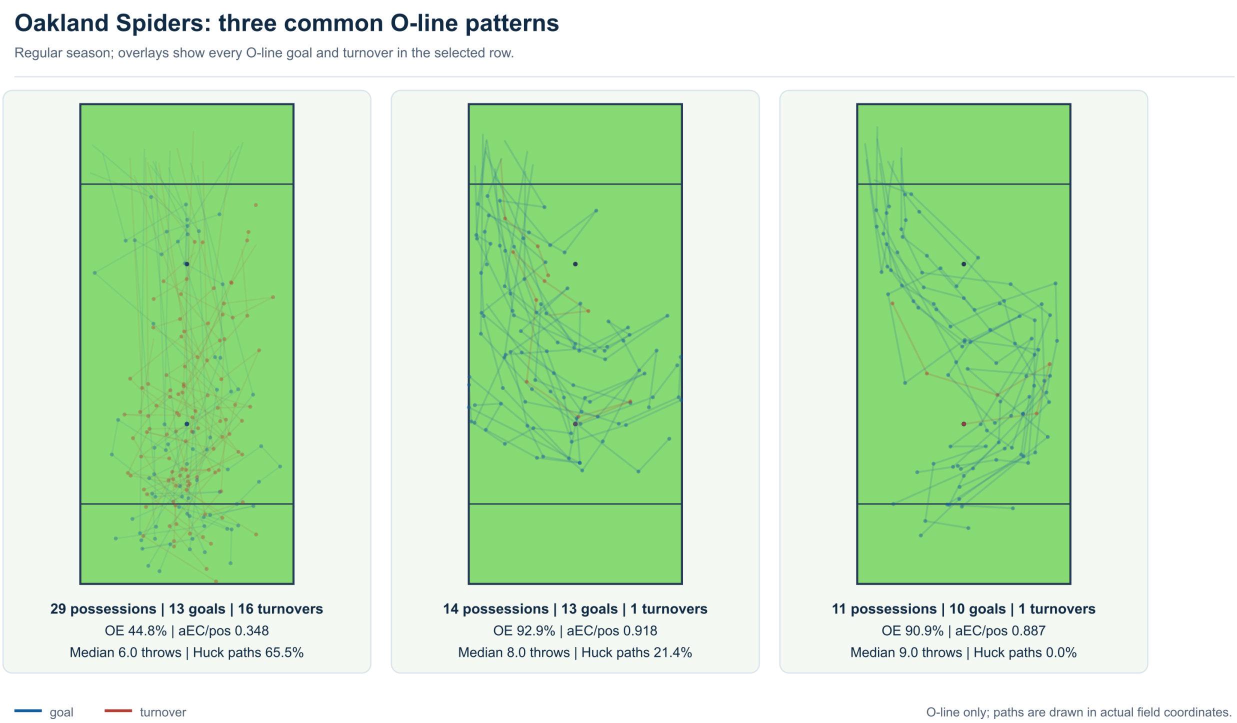

Figure 5: Oakland Spiders' three selected recurring patterns. The first row is the largest and most turnover-heavy selected family; the two smaller rows are much more efficient and lower-huck.

five-throw mode does not prove that a team is trying to reach five throws. It simply identifies the most common observed length in the selected outcome set.

The productive next step is a second manual pass. Once a team has a small set of spatial families, an analyst can name them using both their geometry and their function: for example, a direct deep branch, a lateral continuation sequence, or a long reset-and-advance pattern. The figures make those judgments auditable. Another analyst can see the same overlays, disagree with a boundary, and revise the group without changing the underlying possession reconstruction.

5 Limitations

Several limitations should shape how these results are used.

First, the pattern taxonomy is analyst-curated. The three rows are useful descriptive units, but their boundaries reflect visual judgment and the current arrangement checkpoints. A future version should measure inter-rater agreement or compare the manual rows with a geometry-first clustering baseline.

Second, the paper studies four teams and three selected rows per team. Large unsorted groups remain outside the narrative by design. The figures are not a complete description of any offense, and they should not be used to infer that an unshown pattern is rare without checking the full browser.

Third, the sample is regular season only. It does not answer whether a team changed its offense in the playoffs. Nor does the current summary adjust for opponent pressure, field conditions, score state, personnel, or starting field position. These factors can influence both the path and the outcome.

Finally, the paper uses the cached throw-level aEC values supplied by Shown Space; it does not refit or audit the underlying machine-learning model. The metric is treated as a contextual contribution signal, not as a causal estimate that a pattern alone produced the final score.

6 Conclusion

The four teams in this study do not reduce to a single offensive template. Empire's selected groups combine high conversion with a direct aggressive branch and longer mixed routes. Austin carries a short, aggressive option alongside longer sequences. Minnesota's selected set places a high-density direct branch beside longer mixed paths, with turnovers making the uncertainty visible. Oakland's three rows make the strongest case for heterogeneity: its largest selected family is turnover-heavy, while two smaller families are much more efficient and lower-huck.

The practical result is a way to write about team offense that stays close to the field. The heatmap says how long possessions tend to last. The overlays say where the disc travels. Efficiency and aEC say what those routes produced. Together they turn a possession browser into an argument about how teams build scores, while leaving enough of the evidence visible for the reader to make a different judgment.

Reproducibility

All calculations and figures in this manuscript are regenerated with `scripts/build_paper_figures.py`. The script reads the four saved arrangements, filters the cached game files through July 19, 2026, writes the CSV/LaTeX tables under `paper/generated/`, and produces standalone SVG figures. The arrangement checkpoints remain separate from the generated figures so later manual edits can be compared with this version. The SVG figures are converted to the checked-in PNGs by `scripts/render_paper_figures.cjs` when the figures are regenerated.

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Table 2: Metrics for the three selected patterns per focus team. Efficiency includes turnovers; aEC/pos. is the mean total aEC, and aEC/throw is the aggregate total aEC divided by aggregate throws.

Team	Pat.	n	Share	G	TO	Eff.	aEC/pos.	aEC/throw	Med. throws
New York Empire	1	33	12.8%	27	6	81.8%	0.796	8.0	
New York Empire	2	20	7.8%	16	4	80.0%	0.702	3.0	
New York Empire	3	11	4.3%	9	2	81.8%	0.827	8.0	
Austin Sol	1	25	8.8%	20	5	80.0%	0.762	13.0	
Austin Sol	2	13	4.6%	10	3	76.9%	0.767	10.0	
Austin Sol	3	12	4.2%	9	3	75.0%	0.649	5.0	
Minnesota Wind Chill	1	23	8.6%	16	7	69.6%	0.642	8.0	
Minnesota Wind Chill	2	24	9.0%	19	5	79.2%	0.758	4.5	
Minnesota Wind Chill	3	16	6.0%	11	5	68.8%	0.730	10.5	
Oakland Spiders	1	29	10.9%	13	16	44.8%	0.348	6.0	
Oakland Spiders	2	14	5.3%	13	1	92.9%	0.918	8.0	
Oakland Spiders	3	11	4.1%	10	1	90.9%	0.887	9.0	